

Exercise Sheet **Representation Learning**

Exercise 1: Contrastive Loss (20 P)

Given the SimCLR loss from the lecture for all views i, j from the same samples in a minibatch (MB).

$$\mathcal{L} = -\frac{1}{N} \sum_{i,j \in MB} \log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j) / \tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k) / \tau)} \quad (1)$$

with $\text{sim}(u, v) = \frac{u^T v}{\|u\| \|v\|}$ being the cosine similarity τ a scalar and N the number of samples.

- a) Rewrite the loss explicitly into the following form:

$$\tau \mathcal{L} = \mathcal{L}_a + \mathcal{L}_d$$

with $\mathcal{L}_a = -\frac{1}{N} \sum_{i,j \in MB} \text{sim}(\mathbf{z}_i, \mathbf{z}_j)$.

What is the purpose of $\mathcal{L}_a$ and $\mathcal{L}_d$ in the loss?

- b) How does the parameter τ influences the distance between representations?

Exercise 2: Lecture Questions (20 P)

- a) What is a pretext task? Give four examples for pretext tasks.
- b) What is a representation collapse and how is it prevented in SimCLR?
- c) Given an image/text model with image encoder f and text encoder g which both produce a representation $z \in \mathbb{R}^d$, we want to perform zero-shot classification. Given text labels $t_1, \dots, t_k$ that describe k classes and an image x , how do you compute the predicted class c ?
- d) Name two other applications for representations from a pretext task other than using them for a classification downstream task.

Exercise 3: Programming (60 P)

Download the programming files on ISIS and follow the instructions.